

Comprehensive Report: HealthConnect Healthcare Insights

Executive Summary

This report provides an analysis of patient treatment data to identify critical factors influencing treatment success. Using a decision tree model and demographic analysis, the report highlights actionable insights to improve patient outcomes. Key findings include patterns in treatment success across age groups, gender, and diagnoses, along with recommendations for targeted interventions.

Introduction

HealthConnect is committed to providing innovative healthcare solutions by leveraging data-driven insights. This project analysed a dataset containing patient demographics, diagnoses, treatment plans, and outcomes. The objective was to identify patterns and use predictive analytics to improve treatment success rates.

Data Insights

The dataset provided an overview of 4,999 patients, covering details such as age, gender, diagnosis, treatment plans, and billing amounts. Key insights include:

1. **Age Distribution:** Patients ranged from 0 to 100 years, with a median age of 49.
2. **Gender Breakdown:** A fairly even split between male and female patients.
3. **Treatment Success Rates:** Success rates varied significantly based on treatment type, with medication and lifestyle changes showing higher success compared to surgery.
4. **Billing Amounts:** Billing amounts ranged widely, with a median of \$5,221.

Predictive Model Findings

A decision tree model was developed to predict treatment success based on patient demographics and treatment plans. The model achieved an accuracy of 34%, indicating potential challenges in data quality and feature selection. The confusion matrix revealed balanced but low performance across success categories.

Actionable Recommendations

Based on the findings, the following recommendations are proposed:

1. **Enhanced Data Collection:** Improve the granularity and consistency of patient data, such as recording test results and medical conditions. Also, proper and adequate data entry practices must be ensured.
2. **Personalized Treatment Plans:** Focus on age- and diagnosis-specific interventions to maximise success rates.

3. **Predictive Model Refinement:** Enhance the predictive model by incorporating additional features and addressing class imbalances. This could be more effective if the “Unknown Data” issue is tackled as soon as possible.

4. **Cost Optimization:** Analyze billing patterns to identify cost-effective treatments with high success rates.

Conclusion

The analysis provides a roadmap for improving patient outcomes at HealthConnect. By leveraging data-driven insights and refining predictive models, the organisation can deliver more personalised care and achieve its mission of innovative healthcare solutions. However, we have to pay attention to data quality and it starts from the point of entry.